

Dark Patterns in Indian E-Commerce: Consumer Protection Challenges and Regulatory Gaps

Dr. Ankit Khare¹ Prakriti Khare² Nisha Siddha³

Assistant Professor^{1,2,3}

Baderia Global Institute of Engineering and Management, Jabalpur, Madhya Pradesh, India^{1,2,3}

DOI: <https://doi.org/10.61877/ijmrp.v4i7.327>

Abstract:

India's rapidly expanding e-commerce sector valued at approximately USD 84 billion and projected to grow exponentially has intensified concerns around consumer protection, particularly through the widespread use of *dark patterns*: deceptive user interface (UI) and user experience (UX) designs that exploit cognitive biases to manipulate consumer decision-making. This study undertakes a qualitative, secondary-data-based analysis to examine the prevalence, typologies, and behavioural impacts of dark patterns in Indian e-commerce markets, alongside an evaluation of the existing regulatory response. Drawing on regulatory reports, national audits, consumer surveys, and judicial decisions issued by the Central Consumer Protection Authority (CCPA), Advertising Standards Council of India (ASCI), Local Circles, and allied institutions, the study finds that dark patterns are systemic rather than incidental, with a vast majority of major platforms deploying multiple manipulative design practices. Evidence indicates that consumers with lower income, education, and digital literacy levels are disproportionately vulnerable, exacerbating existing inequalities in India's digital economy. Using nudge sludge theory, the paper demonstrates how platform design strategically activates heuristic-driven, System-1 cognition while obstructing reflective consumer choice, thereby eroding autonomy, informed consent, and fairness. While the Consumer Protection Act, 2019 and the CCPA's 2023 Dark Pattern Guidelines represent important regulatory milestones, enforcement outcomes remain limited, with persistent compliance gaps, weak design-level oversight, and over-reliance on self-regulation. Judicial interventions, signal growing legal recognition of interface-based deception but remain largely reactive. The study concludes that effective consumer protection in digital markets requires a paradigm shift from grievance-centric remedies to preventive design regulation. It recommends statutory design-level prohibitions, mandatory and independent UX audits for dominant platforms, strengthened inter-regulatory coordination, and a unified consumer redressal architecture. By integrating behavioural theory with India-specific empirical and regulatory evidence, this paper contributes to the emerging scholarship on digital consumer protection and offers policy-relevant insights for governing platform-based markets in developing economies.

Keywords: E-commerce, Dark Patterns, Consumer Protection, Nudge Theory, Sludge.

Introduction:

With nearly 900 million internet users, India's e-commerce sector has expanded rapidly, reaching an estimated market size of USD 84 billion in 2024 and projected to exceed USD 350 billion by 2030 (India Brand Equity

Foundation [IBEF], 2024; Statista, 2024). While this growth has significantly enhanced market access and consumer convenience, it has also amplified consumer protection risks arising from manipulative digital practices. Among these, *dark patterns* deceptive user interface (UI) and user experience (UX) design strategies that subvert users' genuine preferences have emerged as a critical regulatory concern in contemporary digital markets (Gray et al., 2018; Mathur et al., 2019). Dark patterns constitute consumer-targeted digital design practices that exploit cognitive biases to steer individuals toward decisions that deviate from their original intentions. Coined by user experience designer Harry Brignull in 2010, the concept has gained increasing scholarly and regulatory attention due to its close association with psychologically manipulative nudges embedded within digital choice architectures. These practices are deeply rooted in human cognition and decision-making processes. As articulated in Daniel Kahneman's dual-process theory, human cognition operates through two interacting systems: *System 1*, which governs fast, automatic, and intuitive responses, and *System 2*, which facilitates slow, deliberative, and reflective reasoning (Kahneman, 2011). Dark patterns are deliberately designed to trigger System 1 processing by leveraging time pressure, default selections, visual salience, and emotional cues, thereby suppressing System 2 engagement. Through the strategic application of behavioural science principles, digital platforms integrate such covert design techniques to optimise conversions, lock-in users, and extract additional value often at the expense of consumers' autonomy, informed consent, and rational choice (Gray et al., 2018; Luguri & Strahilevitz, 2021). Empirical evidence suggests that the prevalence of dark patterns in India is both widespread and sectorally concentrated. In its 2022 discussion paper, the Advertising Standards Council of India (ASCI) identified 13 recurring dark pattern categories in the Indian digital ecosystem, with the fashion (23.2%), e-commerce (16.3%), and cryptocurrency (24.16%) sectors exhibiting the highest incidence of violations (ASCI, 2022). Platform-level assessments indicate that approximately 87% of major digital platforms deploy at least one form of dark pattern, with interface interference (70%) and obstruction-based designs (63%) being the most prevalent (ASCI, 2022; Mathur et al., 2019). From the consumer perspective, nearly 68% of users reported experiencing deceptive online practices, yet only 39% were familiar with the term "dark patterns," revealing a substantial informational asymmetry (Local Circles, 2023). The consequences are tangible: 55% of respondents reported unexpected charges, while 47% experienced difficulty cancelling subscriptions or services, underscoring the structural power imbalance embedded in platform design (Local Circles, 2023). Globally, regulatory attention to dark patterns has intensified. The European Data Protection Board (EDPB), for instance, has issued guidelines identifying manipulative interface practices on social media platforms, including excessive consent prompts, obscured privacy controls, and design choices that nudge users toward data disclosure (EDPB, 2022). As digital business models increasingly rely on behavioural analytics, personal data aggregation, and A/B testing, dark patterns have expanded in both scope and sophistication. In India, however, the term "dark patterns" entered mainstream regulatory discourse relatively late. Prior to statutory intervention, ASCI addressed deceptive design practices through its advertising code, applicable across media, including digital advertising and platform-owned content. Notably, ASCI's 2021 influencer disclosure guidelines sought to address disguised advertising by mandating transparency in paid promotions, highlighting growing concern over blurred

boundaries between content and advertising and their implications for informed consumer choice. Despite rising awareness, India's regulatory response remains fragmented. The Consumer Protection Act, 2019 classifies deceptive digital practices as unfair trade practices; however, enforcement outcomes have been limited, with complaint resolution rates estimated at approximately 28%, despite reported consumer losses exceeding ₹200 crores (Central Consumer Protection Authority [CCPA], 2023; National Consumer Helpline [NCH], 2023). Judicial bodies have begun recognising interface-level deception, as illustrated by the Hyderabad District Consumer Disputes Redressal Commission's ruling against BookMyShow for concealed convenience fees, thereby affirming the applicability of consumer protection law to manipulative digital design (District Consumer Disputes Redressal Commission, Hyderabad, 2023). Sector-specific regulators have also intervened selectively; for example, the Insurance Regulatory and Development Authority of India (IRDAI) prohibited pre-checked travel insurance options in 2020 to curb deceptive default settings (IRDAI, 2020). Nevertheless, a comprehensive, design-centric regulatory framework addressing dark patterns across digital platforms remains underdeveloped (CCPA, 2023). Against this backdrop, the present study addresses three critical gaps in existing scholarship and policy discourse: (i) the absence of a consolidated, India-specific synthesis of dark pattern prevalence based on secondary evidence from regulatory bodies and institutional studies (ASCI, CCPA, NCH); (ii) limited examination of differential consumer vulnerability, particularly among low-income and low-education groups (Luguri & Strahilevitz, 2021); and (iii) insufficient evaluation of regulatory effectiveness following the enactment of the Consumer Protection Act, 2019. Employing qualitative content analysis of regulatory reports, consumer surveys, and empirical platform studies, this research maps dominant dark pattern typologies, analyses consumer harm mechanisms through behavioural frameworks, and proposes targeted regulatory reforms to strengthen consumer protection in India's evolving digital economy.

Objectives of the Paper:

1. To examine the prevalence and dominant typologies of dark patterns deployed across Indian e-commerce platforms using secondary evidence from regulatory reports, institutional studies, and consumer surveys.
2. To analyse the behavioural and cognitive mechanisms through which dark patterns influence consumer decision-making, drawing on dual-process models of cognition and nudge-sludge theory.
3. To evaluate the adequacy and effectiveness of India's existing consumer protection framework, particularly the Consumer Protection Act, 2019, in addressing dark patterns and to identify regulatory gaps requiring policy intervention.

Literature Review:

Conceptualising Dark Patterns in Digital Markets

The concept of *dark patterns* was first introduced by user experience designer Harry Brignull (2010) to describe deceptive interface designs that manipulate users into actions they did not intend to take. Unlike neutral or user-centric design, dark patterns deliberately exploit cognitive biases, information asymmetries, and behavioural heuristics to influence consumer choices in ways that primarily advance platform objectives rather than user welfare (Brignull, 2010; Gray et al., 2018). Subsequent academic scholarship has expanded this

concept by systematically categorising dark patterns and examining their ethical, behavioural, and legal implications within digital environments (Mathur et al., 2019; Luguri & Strahilevitz, 2021). Gray et al. (2018) conceptualise dark patterns as interface designs that subvert user intent through visual manipulation, misleading defaults, and obfuscated choice structures. Empirical investigations by Mathur et al. (2019) further demonstrate that such practices are pervasive across online platforms, including e-commerce marketplaces, subscription-based services, and mobile applications. These studies emphasise that dark patterns are not accidental design flaws but intentionally embedded strategies within digital choice architectures aimed at maximising engagement, retention, and revenue extraction.

Typologies of Dark Patterns

The literature identifies multiple typologies of dark patterns based on their functional characteristics and behavioural effects. Commonly documented practices include *drip pricing*, *confirm-shaming*, *roach motels*, *forced action*, *basket sneaking*, and *disguised advertisements* (Brignull, 2010; Mathur et al., 2019). Human-computer interaction (HCI) research has further grouped these practices into broader categories such as *interface interference*, *obstruction*, *forced action*, and *deceptive framing*, providing a structured framework for empirical analysis and regulatory assessment (Gray et al., 2018). Regulatory bodies have increasingly adopted these typologies to guide enforcement and compliance. In the Indian context, the Advertising Standards Council of India (ASCI) identified thirteen recurring dark pattern practices prevalent across digital advertising and e-commerce platforms in its 2022 discussion paper (ASCI, 2022). These classifications closely align with global academic literature, indicating that dark pattern manifestations are largely consistent across jurisdictions, notwithstanding differences in regulatory maturity and enforcement capacity.

Behavioural Foundations: Nudge Theory and Dual-Process Cognition

Dark patterns are deeply grounded in behavioural economics and cognitive psychology. *Nudge theory*, popularised by Thaler and Sunstein (2008), explains how subtle modifications in choice architecture can influence decision-making without eliminating formal freedom of choice. While nudges are often framed as welfare-enhancing interventions, subsequent scholarship cautions that similar techniques can be strategically misused in commercial contexts to prioritise firm interests over consumer autonomy (Luguri & Strahilevitz, 2021). Kahneman's (2011) dual-process theory provides a cognitive explanation for the effectiveness of dark patterns. Human decision-making operates through two interacting systems: *System 1*, which is fast, intuitive, and automatic, and *System 2*, which is slow, deliberative, and analytical. Dark patterns predominantly target *System 1* processing by exploiting heuristics such as default bias, loss aversion, scarcity effects, and social pressure (Kahneman, 2011; Gray et al., 2018). By activating these heuristics, dark patterns suppress reflective evaluation and significantly reduce consumers' capacity to make informed and autonomous choices.

Dark Patterns as Sludge

Recent literature introduces the concept of *sludge* to describe choice architectures that impose friction, complexity, or cognitive burden, thereby impeding optimal decision-making (Sunstein, 2021). Unlike nudges, which are designed to facilitate beneficial outcomes, sludge operates by increasing effort costs, exploiting inertia, and discouraging protective or corrective actions. Several scholars argue that dark patterns represent a

commercial manifestation of sludge (Luguri & Strahilevitz, 2021; Sunstein, 2021). Practices such as *roach motels*, *obstruction*, and *subscription traps* deliberately make consumer-protective actions such as cancellations, refunds, or comparison shopping difficult and time-consuming. Empirical studies suggest that sludge-based designs disproportionately affect consumers with limited digital literacy, time constraints, or cognitive resources, raising broader concerns about fairness, distributive justice, and structural inequality in digital markets (Mathur et al., 2019).

Consumer Harm and Vulnerability

Existing scholarship consistently documents the adverse effects of dark patterns on consumer welfare. Identified harms include financial losses arising from hidden charges and unintended subscriptions, erosion of consumer trust, diminished transparency, and the undermining of consumer autonomy (Gray et al., 2018; Mathur et al., 2019). Survey-based research further indicates that while a majority of users encounter manipulative interface designs, awareness of the concept of dark patterns remains limited, exacerbating information asymmetries between platforms and consumers (Local Circles, 2023). The literature also highlights that vulnerability to dark patterns is unevenly distributed across socio-economic groups. Consumers with lower income levels, limited education, and reduced digital literacy are significantly more susceptible to deceptive design practices and face greater difficulty navigating complex digital interfaces (Luguri & Strahilevitz, 2021). In emerging digital economies such as India where rapid internet adoption has outpaced investments in digital literacy these vulnerabilities are particularly pronounced, reinforcing the need for robust regulatory intervention and design-level safeguards.

Regulatory Approaches and Gaps

Globally, regulatory responses to dark patterns remain fragmented and unevenly enforced. While jurisdictions such as the European Union and the United States have begun recognising dark patterns within consumer protection and data protection frameworks, regulatory interventions continue to rely heavily on disclosure-based and consent-oriented standards (European Data Protection Board [EDPB], 2022; Federal Trade Commission [FTC], 2022). Such approaches are increasingly criticised for their limited effectiveness in addressing behavioural manipulation embedded within interface design, particularly where cognitive biases are deliberately exploited to override user autonomy (Gray et al., 2018; Sunstein, 2021). In the Indian context, the Consumer Protection Act, 2019 (CPA) provides a broad statutory framework to address unfair trade practices, including deceptive and misleading digital conduct. Recent advisories and enforcement actions by the Central Consumer Protection Authority (CCPA) indicate growing regulatory recognition of dark patterns as a distinct category of consumer harm (CCPA, 2023). However, the literature identifies substantial regulatory gaps, particularly in relation to design-level prohibitions, proactive monitoring mechanisms, and standardised compliance benchmarks for digital platforms (ASCI, 2022; Luguri & Strahilevitz, 2021). Judicial interventions—such as penalties imposed for concealed fees and sector-specific bans, including restrictions on pre-selected defaults in insurance products, represent important developments but remain largely reactive and case-specific rather than preventive and systemic (IRDAI, 2020; District Consumer Disputes Redressal

Commission, Hyderabad, 2023). Consequently, enforcement continues to lag behind the scale and sophistication of manipulative design practices in India's rapidly expanding digital markets.

Research Gap:

Despite growing international scholarship on dark patterns, India-specific research that integrates behavioural theory, empirical evidence, and regulatory analysis remains limited. Existing studies tend to focus either on design typologies and interface classifications or on consumer perceptions and awareness, with comparatively little attention devoted to regulatory effectiveness, enforcement outcomes, and the differential impact of dark patterns on vulnerable consumer groups (Mathur et al., 2019; Luguri & Strahilevitz, 2021). Moreover, there is a paucity of work examining how behavioural manipulation intersects with institutional capacity constraints in emerging digital economies. This study addresses these gaps by synthesising secondary regulatory data, institutional reports, and consumer survey evidence through the analytical lens of dual-process cognition and nudge sludge theory. By situating dark patterns within India's specific legal and market context, the research offers a contextualised assessment of consumer protection challenges in the e-commerce sector and contributes policy-relevant insights into strengthening design-level regulation and enforcement mechanisms.

Research Methodology:

This study adopts a qualitative, descriptive, and analytical research design, relying exclusively on secondary data to examine the prevalence, behavioural implications, and regulatory treatment of dark patterns in India's e-commerce ecosystem. A doctrinal empirical hybrid approach is employed, combining qualitative content analysis of regulatory and institutional documents with synthesis of consumer survey evidence. This methodological design is appropriate for analysing complex digital market practices that operate at the intersection of law, behavioural science, and platform governance, where primary experimentation or access to proprietary platform data is limited (Gray et al., 2018; Luguri & Strahilevitz, 2021). The study does not seek to establish causal relationships but instead aims to map patterns, interpret behavioural mechanisms, and evaluate regulatory adequacy within an emerging digital economy context. Such an approach is consistent with prior interdisciplinary research on dark patterns and digital consumer protection (Mathur et al., 2019).

Data Sources:

Secondary data were collected from authoritative, publicly available sources to ensure reliability, transparency, and policy relevance. The principal categories of data sources are as follows:

(a) Regulatory and Institutional Reports

- Advertising Standards Council of India (ASCI) discussion papers and compliance reports
- Central Consumer Protection Authority (CCPA) guidelines, advisories, and enforcement records
- National Consumer Helpline (NCH) complaint statistics and annual summaries

(b) Consumer Survey Reports

- Local Circles national consumer perception surveys on e-commerce practices
- Government- and industry-sponsored studies on digital consumer behaviour and online marketplaces

(c) Judicial and Regulatory Documents

- Orders of Consumer Disputes Redressal Commissions (e.g., the *BookMyShow* case)

- Sector-specific regulatory circulars, including the Insurance Regulatory and Development Authority of India (IRDAI) prohibition on pre-selected insurance options

(d) Academic Literature

- Peer-reviewed journal articles on dark patterns, behavioural manipulation, nudge–sludge theory, and human–computer interaction (HCI) taxonomies

These sources collectively enable triangulation between regulatory intent, empirical evidence of platform practices, and documented consumer harm.

Analytical Framework and Method

A qualitative content analysis technique was employed to systematically review regulatory documents, institutional reports, and judicial decisions. Dark pattern practices were coded and categorised using established typologies from HCI and behavioural economics literature, including classifications proposed by Gray et al. (2018) and Mathur et al. (2019). Consumer harm was analysed through behavioural lenses, particularly dual-process cognition and nudge sludge theory, to assess how interface designs influence decision-making and constrain consumer autonomy. Survey findings reported by institutional sources were synthesised descriptively to identify recurring trends in consumer experience, awareness, and vulnerability. The analysis focused on convergence across sources rather than isolated findings, thereby strengthening interpretive validity.

Sampling Framework:

For platform-level analysis, the study draws on empirical findings reported in prior audits and institutional studies covering approximately 30 major e-commerce and digital service platforms operating in India. These include online marketplaces, travel and ticketing portals, and subscription-based digital services. Platform selection in the underlying studies was based on criteria such as market share, size of user base, and regulatory relevance, ensuring coverage of dominant actors within the Indian digital economy. Consumer vulnerability analysis is derived from aggregated survey data reported by institutional sources. Key variables examined include income level, educational attainment, awareness of dark patterns, and reported experiences of financial or procedural harm. Given the secondary nature of the data, the study does not claim population-wide generalisability but provides a robust descriptive and analytical account of systemic trends affecting Indian consumers.

Analysis and Discussion

Analysis of secondary institutional reports, regulatory audits, and empirical platform studies reveals the systemic prevalence of dark patterns across Indian e-commerce and digital service platforms. Synthesised evidence indicates that approximately 87% of major platforms operating in India deploy at least one form of dark pattern, demonstrating that manipulative interface design is not an isolated deviation but a structural feature of platform-based commerce

Typologies of Dark Patterns, Cognitive Biases, and Consumer Harm

Table 1 presents a consolidated typology of dark patterns prevalent in Indian e-commerce, mapping each practice to its underlying System 1 cognitive bias and associated consumer harm. The analysis illustrates how

platforms systematically exploit fast, heuristic-driven decision-making to influence user behaviour, consistent with Kahneman's (2011) dual-process model.

Table 1: Dark Patterns in Indian E-Commerce: Typology, Cognitive Biases, Evidence and Consumer Harm

Dark Pattern Brignull (2010)	Description	System-1 Bias Exploited As per Daniel Kahneman's dual-process theory (2011)	Evidence in India	Observed Consumer Harm
Drip Pricing	Incremental disclosure of mandatory or optional charges during the checkout process prevents accurate price comparison at the outset.	Anchoring bias; Loss aversion	Widely reported in travel and ticketing platforms (ASCI, 2022); 55% consumers faced surprise charges	Unexpected financial burden; distorted price perception
Confirm-Shaming	Use of guilt-inducing language to frame opt-out choices negatively, pressuring users to comply.	Social pressure; Affect heuristic	Frequently observed in subscription prompts (ASCI, 2022)	Coerced consent; emotional manipulation
Roach Motel	Easy sign-up processes combined with intentionally difficult cancellation or exit mechanisms.	Status quo bias; Inertia	47% users reported difficulty cancelling services (LocalCircles, 2023)	Forced continuation of paid services
Forced Action	Requiring users to complete unrelated actions (e.g., account creation, data sharing) to access services.	Compliance bias	Common across marketplaces and apps (CCPA, 2023)	Loss of autonomy; unnecessary data disclosure
Interface Interference	Manipulation of visual hierarchy (colour, size, placement) to privilege platform-preferred choices.	Salience bias; Visual dominance	Identified in ~70% of audited platforms	Skewed decision-making; misleading consent
Obstruction	Introducing friction, complexity, or delays to discourage protective actions such as refunds or cancellations.	Cognitive fatigue; Friction cost	Present in ~63% of platforms analysed	Consumer frustration; financial loss
Basket Sneaking	Addition of extra items or services to carts without explicit user consent.	Default bias	Common in insurance and add-on services	Unintended purchases
Disguised Advertisements	Ads designed to resemble editorial or user-generated content.	Source confusion; Trust heuristic	High incidence in fashion & crypto sectors	Misleading information; reduced trust
False Urgency / Scarcity	Misleading countdown timers or stock alerts creating artificial pressure.	Scarcity effect; Time pressure	Frequently documented by ASCI	Impulsive purchases
Pre-Selected Defaults	Automatically selected options favouring the platform (e.g., add-ons, insurance).	Default bias	Banned by IRDAI in insurance (2020)	Uninformed consent; financial leakage
Nagging	Repeated prompts urging users to take a specific action after refusal.	Wear-down effect	Common in app notifications	Decision fatigue
Hidden Information	Concealment of material information within dense text or obscure links.	Information overload	Identified across e-commerce disclosures	Lack of informed choice
Subscription Traps	A combination of default enrolment, unclear billing, and complex exit processes.	Inertia; Loss aversion	Increasing consumer complaints (NCH, 2023)	Recurring unwanted payments

Source: (Data compiled from ASCI, 2023; CCPA, 2023; Jain & Vyas, 2025)

Above Table demonstrates that dark patterns such as drip pricing, interface interference, obstruction, and subscription traps are particularly pervasive. These practices impair informed price comparison, restrict exit

options, and distort consent mechanisms, resulting in tangible consumer harm including unexpected financial costs, coerced continuation of services, and erosion of autonomy (ASCI, 2023; Local Circles, 2023). Importantly, these patterns are not merely deceptive but behaviourally engineered, exploiting anchoring, default bias, loss aversion, and cognitive fatigue. Content analysis further reveals that interface interference ($\approx 70\%$) and obstruction-based designs ($\approx 63\%$) are the most frequently deployed categories, indicating a strategic emphasis on manipulating choice architecture rather than overt misinformation. This finding reinforces global literature suggesting that digital manipulation increasingly operates through design rather than content (Gray et al., 2018; Mathur et al., 2019). Sectoral concentration of violations most notably in cryptocurrency (24.16%), fashion (23.2%), and e-commerce marketplaces (16.3%) reflects heightened commercial incentives to deploy aggressive conversion-oriented designs in high-competition, high-margin environments (ASCI, 2023).

Consumer-Level Prevalence and Behavioural Impact

Table 2 : Prevalence of Key Dark Patterns in Indian E-Commerce Platforms

Dark Pattern Identified	% of Consumers Affected	Common Manifestation	Primary Consumer Harm
Drip Pricing	75%	Hidden platform, payment handling, or COD fees revealed at checkout	Financial loss; distorted price comparison
Bait and Switch	48%	Price or offer changes after login or during checkout	Misleading expectations; coerced upgrades
Privacy Zuckering	44%	Use of personal data without explicit consent for targeting	Privacy erosion; loss of informational control
Forced Action	29%	Orders processed via COD or Pay Later despite cancellation	Loss of autonomy; unwanted transactions
Basket Sneaking	21%	Auto-added installation services or donations	Unintended purchases; billing opacity

Source: LocalCircles (2025). National audit on dark patterns in Indian e-commerce platforms (June–September 2025)

Table 2 synthesises findings from the **Local Circles (2025)** national audit, by linking dark pattern prevalence to consumer experience. Drip pricing emerged as the most prevalent practice, affecting **75% of consumers**, followed by **bait-and-switch (48%)** and **privacy zuckering (44%)**. These patterns distort expectations formed during browsing and undermine informational self-determination, particularly through non-consensual data exploitation. Forced action (29%) and basket sneaking (21%) further demonstrate how platforms leverage procedural dominance to override consumer intent. Collectively, these findings confirm that **dark patterns operate across multiple stages of the consumer journey**, from discovery to checkout to post-purchase engagement, magnifying cumulative harm.

Platform-Level Evidence and Festive Sale Amplification

Table 3: Evidence of Dark Patterns During Festive Sales on Indian E-Commerce Platforms

Platform / Event	Festive Sale Context	Dark Patterns Identified	Illustrative Evidence	Consumer Impact
Flipkart	Big Billion Days 2025	False urgency; Drip pricing; Bait and switch	Heavily promoted smartphone discounts cancelled post-payment or offered minimal real price reduction	Misleading price expectations; delayed refunds
Amazon	Great Indian	Drip pricing; Basket	Checkout-stage fee disclosures;	Financial loss;

	Festival 2025	sneaking; Subscription nudging	add-on protection plans pushed during purchase flow	coerced add-on purchases
Multiple Platforms	Festive flash sales	False scarcity; Premium membership nudging	Countdown timers and “limited stock” alerts without verifiable scarcity	Impulsive purchases; distorted choice
Cross-platform trend	Peak sales period	Interface interference; Obstruction	Complex refund workflows and delayed reversals despite successful payment	Consumer frustration; trust erosion

Source: LocalCircles (2025). *National audit on dark patterns in Indian e-commerce platforms (June–September 2025)*

During India’s 2025 festive sales, dark patterns were especially pronounced, acting as a “stress test” for manipulative online marketplace practices (MediaNama, 2025). Platforms hyped deep discounts and urgency messaging, but consumers frequently encountered cancellations after confirmed orders, hidden checkout charges, and misleading discount claims (MediaNama, 2025). For example, Flipkart’s Big Billion Days sale advertised iPhone deals that were either cancelled after payment or offered negligible actual discounts when compared with manufacturer prices. Similarly, Amazon’s Great Indian Festival promotions were associated with delayed refunds and order cancellations despite payment settlements (MediaNama, 2025). The audit-style checks confirmed the presence of classic dark patterns such as drip pricing with undisclosed fees appearing only at the checkout and basket sneaking, where add-on protection plans were pushed during purchase flow. Platforms also engaged in false urgency via countdown timers and premium membership upsells that failed to deliver promised benefits (MediaNama, 2025). These observations suggest that self-audit declarations by platforms, as mandated by the Central Consumer Protection Authority’s 2025 advisory, have had limited impact on actual design practices, particularly during peak sales events (MediaNama, 2025).

Nudge Sludge Interpretation and Ethical Implications

Table 4: Examples of Dark Patterns in Indian E-Commerce Platforms

Dark Pattern	Prevalence (%)	Examples
False urgency	42	“Only 2 rooms left! 30 others viewing now”
Basket sneaking	35	Auto-added insurance with flight tickets
Confirm-shaming	28	“I will stay unsecured” phrase for insurance
Forced action	22	Mandatory newsletter signup
Subscription trap	31	Difficult cancellation of free trials
Interface interference	40	Misleading cancel/close buttons
Bait and switch	25	Out-of-stock switch to higher-priced item
Drip pricing	38	Price updated at checkout
Disguised ads	30	Ads styled as user content
Nagging	33	Repeated notifications

Source: (Data compiled from ASCI, 2023; CCPA, 2023; Jain & Vyas, 2025)

The distribution patterns in **Table 4** align closely with **nudge-sludge theory** (Thaler & Sunstein, 2008; Sunstein, 2021). High-prevalence practices such as **false urgency, drip pricing, and nagging** operate as

exploitative nudges by activating **System 1 cognition**, suppressing deliberation and inflating impulsive decision-making. In contrast, **subscription traps, obstruction, and forced action** function as **sludge**, deliberately increasing friction to discourage consumer-protective actions such as cancellation or refund requests. Critically, the near parity between nudge-based (42–38%) and sludge-based (40–31%) patterns indicates a **shift from persuasion to coercion** in digital choice architectures. This convergence erodes the normative legitimacy of nudging and raises profound ethical concerns regarding **consent, autonomy, and procedural fairness** in algorithmically mediated markets.

Regulatory Effectiveness and Enforcement Gaps

The first formal attempt in India to identify and enforce self-regulation against dark patterns occurred in **June 2023**, when the Advertising Standards Council of India (ASCI) issued the *Guidelines for Online Deceptive Design Patterns in Advertising* (ASCI, 2023). These guidelines targeted four types of dark patterns in advertisements; however, as ASCI functions as a self-regulatory body, it lacked enforceable powers (SCC Times, 2025). Recognizing the widespread use of manipulative tactics on digital platforms, the **Central Consumer Protection Authority (CCPA)** established under Section 104 of the Consumer Protection Act, 2019 exercised its powers under Section 186 of the Act to issue the *Guidelines for Prevention and Regulation of Dark Patterns, 2023* (CCPA, 2023; SCC Times, 2025). These CCPA guidelines formally defined dark patterns and enumerated **13 inclusive categories**, including false urgency, basket sneaking, subscription traps, nagging, and drip pricing (SCC Times, 2025). These patterns represent the manipulative design strategies that many consumers encounter daily, such as artificially induced scarcity, automatic addition of products to carts, difficulty unsubscribing from services, repetitive notifications, and hidden or incrementally disclosed charges during checkout. On National Consumer Day, December 24, 2024, the Government of India launched three digital tools to enhance dark pattern awareness and enforcement: the Jago Grahak Jago App, e-Jagriti App, and Jagriti Dashboard. The e-Jagriti App digitizes Consumer Disputes Redressal Commissions nationwide, enabling consumers to file complaints, access rights information, and track case status seamlessly. The Jago Grahak Jago App consolidates consumer services on a single platform, featuring dark pattern identification tools. The Jagriti Dashboard provides real-time transparency in dispute resolution processes (SCC Online, 2025). However, these initiatives confront fundamental structural limitations that undermine their practical effectiveness. Their efficacy critically depends on consumer digital literacy and technological proficiency precisely the demographics most vulnerable to dark patterns, including those with lower education levels and residents of Tier-II/III cities (Chugh & Jain, 2021). This creates an inherent access paradox where digital solutions target the digitally illiterate, inadvertently exacerbating existing inequalities. The deployment of three separate applications for interconnected functions exemplifies design inefficiency, requiring consumers to navigate multiple downloads, logins, and interfaces rather than providing true single-window redressal. While these tools facilitate complaint filing and status tracking, they fail to deliver comprehensive dispute resolution, lacking essential features such as virtual hearings, evidence upload capabilities, and final resolution delivery (SCC Online, 2025). This represents a partial solution addressing access but ignoring substantive justice delivery. Empirical evidence underscores limited impact

despite statutory recognition and grievance infrastructure: national complaint resolution rates remain stagnant at 28%, 81% of platforms falsely declared themselves "dark pattern free" during CCPA self-audits, and consumer awareness of recourse mechanisms remains minimal (LocalCircles, 2025). The implementation failure is evident: guidelines and apps do not equate to enforcement, nor does awareness plus filing guarantee relief. Policy recalibration requires a three-phase approach. Phase 1 demands a Unified Consumer Protection App integrating all functions with mandatory platform API connectivity and offline USSD/SMS access for non-smartphone users. Phase 2 shifts focus to prevention through bright-line prohibitions against specific patterns, mandatory UX design audits for dominant platforms, and algorithmic transparency requirements. Phase 3 emphasizes capacity building through targeted consumer education campaigns in vulnerable regions and specialized judicial training for digital evidence adjudication. Ultimately, while the intent behind these digital intervention merits commendation, their translation into effective consumer protection remains constrained. Technology serves as an essential augmentation but cannot substitute preventive design regulation targeting sludge architecture at its source (Thaler, 2018). The fundamental policy shift must transfer liability from fragmented consumer complaints to platform accountability through systematic manipulation detection and prohibition. On June 5, 2025, the Central Consumer Protection Authority (CCPA) issued a pivotal advisory under Section 18(1) of the Consumer Protection Act, 2019, mandating e-commerce platforms to conduct internal audits for dark pattern detection within three months (Department of Consumer Affairs, 2025a). The directive explicitly references the Guidelines for Prevention and Regulation of Dark Patterns, 2023, and Rule 4(9) of the Consumer Protection (E-Commerce) Rules, 2020, which mandates clear and affirmative consent through explicit user action rather than pre-checked defaults or implied agreement.

The advisory establishes a dual compliance mechanism:

- Mandatory internal audits to systematically identify manipulative design practices
- Voluntary self-declarations inviting platforms to affirm dark pattern-free status

Concurrently, the Department of Consumer Affairs constituted a Joint Working Group through a June 5 memorandum, tasked with regular meetings over 12 months to monitor compliance and submit progress reports (Department of Consumer Affairs, 2025b). In November 2025, the Union Ministry of Consumer Affairs, Food and Public Distribution reported that 26 leading e-commerce and digital platforms submitted voluntary self-declarations confirming compliance with the *Guidelines for Prevention and Regulation of Dark Patterns, 2023* (expressing agreement with measures to eliminate manipulative UI/UX) after conducting internal or third-party audits to identify and address dark pattern practices. The platforms span diverse sectors including retail, travel, food delivery, healthcare, and general e-commerce. Companies such as Flipkart, Myntra, Zomato, Swiggy, BigBasket, Tata 1mg, JioMart, Reliance Digital, Zepto, Cleartrip, and Meesho reported that audits found either no dark patterns or that previously identified issues have been addressed, with Meesho specifically affirming adherence to all 13 identified dark patterns and maintaining ongoing monitoring. According to the press release, these companies conducted either internal self-audits or

third-party reviews to identify, assess, and eliminate dark patterns, and all declared that their platforms are free from manipulative user interface designs that could mislead or manipulate consumer decision-making. The CCPA appreciated this industry-wide compliance, describing it as exemplary and encouraging other digital service providers to adopt similar self-regulatory measures (Press Information Bureau, 2025).

Policy Recommendations

To effectively combat dark patterns in India's digital marketplace, policymakers must implement a comprehensive, multi-pronged strategy that shifts from reactive grievance redressal to proactive design prevention. First, the Consumer Protection Act, 2019 requires statutory strengthening through bright-line prohibitions converting the 2023 Dark Pattern Guidelines into enforceable law, while extending Central Consumer Protection Authority (CCPA) powers including compulsory UI/UX audits and meaningful penalties to State and National Consumer Disputes Redressal Commissions. Rather than handling individual complaints, CCPA should focus on systemic platform investigations, supported by dynamic updates to banned pattern registries and nationwide consumer awareness campaigns targeting vulnerable demographics. Second, CCPA operationalization as a central digital enforcement agency demands inter-regulatory coordination with RBI, IRDAI, TRAI, and the Data Protection Board through formal MoUs, establishing unified national standards for digital consumer protection. This agency-led approach prioritizes AI-powered platform audits over fragmented consumer complaints, ensuring consistent enforcement across sectors. Third, platforms must face statutory ethical design mandates incorporating privacy-by-design principles, transparent algorithmic governance, and consumer-first defaults, backed by randomized UI/UX audits quarterly for top 20 platforms, semi-annually for the next 30 with violations triggering immediate operational freezes until remediation. Ethical platforms should receive regulatory fast-tracking and public certification as market differentiators. Fourth, consumers need a single-window redressal ecosystem through a unified national portal consolidating pattern reporting, case tracking, virtual hearings, multilingual support, and public enforcement transparency, with mandatory 90-day API integration for all platforms. Finally, sector regulators and CCPA must collaborate on large-scale awareness campaigns while promoting ethical design through academic-industry partnerships and annual recognition awards. This phased implementation statutory reform (0-6 months), institutional restructuring (6-18 months), full enforcement (18-36 months) transforms India's digital marketplace from arena of manipulation to domain of informed choice, ensuring consumer welfare drives technological innovation rather than exploitation.

Conclusion:

This study examined the prevalence, behavioural foundations, and regulatory treatment of dark patterns in Indian e-commerce through a synthesis of secondary regulatory data, consumer surveys, platform audits, and institutional reports. The findings demonstrate that dark patterns are not isolated design anomalies but a **systemic feature of India's digital marketplace**, embedded within platform architectures to influence consumer behaviour at multiple decision points. Despite increasing regulatory recognition, manipulative interface designs continue to persist across platforms, sectors, and sales contexts. Drawing on **dual-process theory and nudge-sludge frameworks**, the analysis shows that dark patterns function by deliberately

activating fast, heuristic-driven System 1 cognition while simultaneously obstructing reflective System 2 decision-making. High-prevalence practices such as drip pricing, false urgency, and confirm-shaming operate as exploitative nudges, whereas obstruction, subscription traps, and forced action exemplify sludge by introducing artificial friction that discourages consumer-protective behaviour. The near parity between nudge-based and sludge-based patterns signals a concerning shift from persuasion to coercion in digital choice architecture. Empirical evidence further reveals a significant **regulatory–enforcement gap**. Although the Consumer Protection Act, 2019 and the CCPA’s 2023 Dark Pattern Guidelines represent an important doctrinal advancement, compliance remains largely declaratory. National audits indicate that a vast majority of platforms continue to deploy dark patterns despite self-audit requirements and voluntary declarations. Digital grievance mechanisms, while improving access, remain fragmented and dependent on consumer awareness and digital literacy precisely where vulnerability is highest. The study concludes that **reactive, complaint-driven regulation is structurally inadequate** for addressing design-based manipulation. Effective consumer protection in digital markets requires a preventive, design-centric regulatory approach that shifts accountability from individual consumers to platforms. Bright-line prohibitions, mandatory UX audits, algorithmic transparency, and unified enforcement mechanisms are essential to counteract behavioural exploitation at scale. By integrating behavioural theory with Indian regulatory evidence, this research contributes to emerging scholarship on digital consumer protection in the Global South. It underscores the need for **institutional capacity-building, design-level regulation, and behavioural awareness** to ensure that technological innovation advances consumer welfare rather than undermines autonomy and informed choice. Future research should empirically examine firm-level compliance post-2025 and assess the long-term effectiveness of preventive design regulation in platform governance.

References:

- Advertising Standards Council of India. (2022). Dark patterns in advertising: A discussion paper. ASCI.
- Advertising Standards Council of India. (2023). Guidelines for online deceptive design patterns in advertising. ASCI.
- Brignull, H. (2010). Dark patterns: Deception vs. honesty in UI design. <https://www.darkpatterns.org>
- Central Consumer Protection Authority. (2023). Guidelines for prevention and regulation of dark patterns, 2023. Ministry of Consumer Affairs, Government of India.
- Chugh, R., & Jain, M. (2021). Digital literacy and consumer vulnerability in India’s platform economy. *Journal of Consumer Policy*, 44(3), 421–439. <https://doi.org/10.1007/s10603-021-09489-1>
- Department of Consumer Affairs. (2025a, June 5). Advisory to e-commerce platforms regarding self-audit for dark patterns. Ministry of Consumer Affairs, Food and Public Distribution.
- Department of Consumer Affairs. (2025b, June 5). Office memorandum on constitution of joint working group on dark patterns. Ministry of Consumer Affairs, Food and Public Distribution.
- District Consumer Disputes Redressal Commission, Hyderabad. (2023). *BookMyShow Pvt. Ltd. v. Consumer complaint order*. Government of Telangana.

- Gray, C. M., Kou, Y., Battles, B., Hoggatt, J., & Toombs, A. (2018). The dark (patterns) side of UX design. *Proceedings of the ACM on Human-Computer Interaction*, 2(CSCW), Article 138. <https://doi.org/10.1145/3274309>
- Insurance Regulatory and Development Authority of India. (2020). Circular on prohibition of pre-checked insurance options in digital sales. IRDAI.
- Internet and Mobile Association of India. (2024). India e-commerce market outlook. IAMAI.
- Jain, R., & Vyas, A. (2025). Dark patterns and consumer autonomy in Indian digital markets. *Indian Journal of Law and Technology*, 21(1), 55–78.
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- LocalCircles. (2023). Dark patterns and deceptive practices in Indian e-commerce: Consumer survey report. LocalCircles Foundation.
- LocalCircles. (2025). National audit on dark patterns in Indian e-commerce platforms (June–September 2025). LocalCircles Foundation.
- Luguri, J. B., & Strahilevitz, L. J. (2021). Shining a light on dark patterns. *Journal of Legal Analysis*, 13(1), 43–109. <https://doi.org/10.1093/jla/laaa006>
- Mathur, A., Acar, G., Friedman, M. J., Lucherini, E., Mayer, J., Chetty, M., & Narayanan, A. (2019). Dark patterns at scale: Findings from a crawl of 11K shopping websites. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW), Article 81. <https://doi.org/10.1145/3359183>
- MediaNama. (2025, June 10). Govt forms task force to curb dark patterns in apps, calls on platforms to self-audit. <https://www.medianama.com/2025/06/223-govt-task-force-dark-patterns-apps/>
- MediaNama. (2025, October 9). LocalCircles audit: 97% e-commerce platforms still use dark patterns; 75% users face hidden fees. <https://www.medianama.com/2025/10/223-localcircles-e-commerce-platforms-dark-patterns-hidden-fees/>
- MediaNama. (2025, September 26). Dark patterns cloud India's festive sales on e-commerce platforms: What it means. <https://www.medianama.com/2025/09/223-dark-patterns-indias-festive-sales-e-commerce-platforms/>
- National Consumer Helpline. (2023). Annual consumer complaint statistics. Department of Consumer Affairs, Government of India.
- Press Information Bureau. (2025, November 20). 26 leading e-commerce platforms declare compliance with self-audit to eliminate dark patterns. Ministry of Consumer Affairs, Food and Public Distribution. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2191948®=3&lang=2>
- Statista. (2024). E-commerce market size in India from 2018 to 2030. Statista Research Department.
- Sunstein, C. R. (2021). *Sludge: What stops us from getting things done and what to do about it*. MIT Press.
- Thaler, R. H. (2018). Nudge, not sludge. *Science*, 361(6401), 431. <https://doi.org/10.1126/science.aau9241>
- Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving decisions about health, wealth, and happiness*. Yale University Press.

The Hindu. (2025, November). E-commerce firms say compliant with dark pattern guidelines. The Hindu.
<https://www.thehindu.com/business/e-commerce-firms-say-compliant-with-dark-pattern-guidelines/article70304337.ece>